

DATA LOGGING Using microSD And RASPBERRY PI PICO

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This data logging system can be used to interface an external or internal sensor of Raspberry Pi Pico board. The project has many advantages, including the use of internal temperature sensor of Raspberry Pi Pico board; so an external temperature sensor is not required. Second, the easily detachable microSD card can be used to record different temperature data at a particular place or industrial setup. The SD card can be removed and carried to laboratory or control room for analysis or future reference. The project also includes optional external sensors, such as force and light sensors.

Components required for the project are given in the Table below.

The author's prototype wired on the breadboard is shown in Fig. 1 and the connection of data logging system with microSD card and Raspberry Pi Pico is shown in Fig. 2. An LDR or any other suitable analogue sensor, such as the force sensor shown, may be used.

The setup uses serial peripheral interface (SPI) pins MOSI, MISO, CS, and CLK of Raspberry Pi Pico

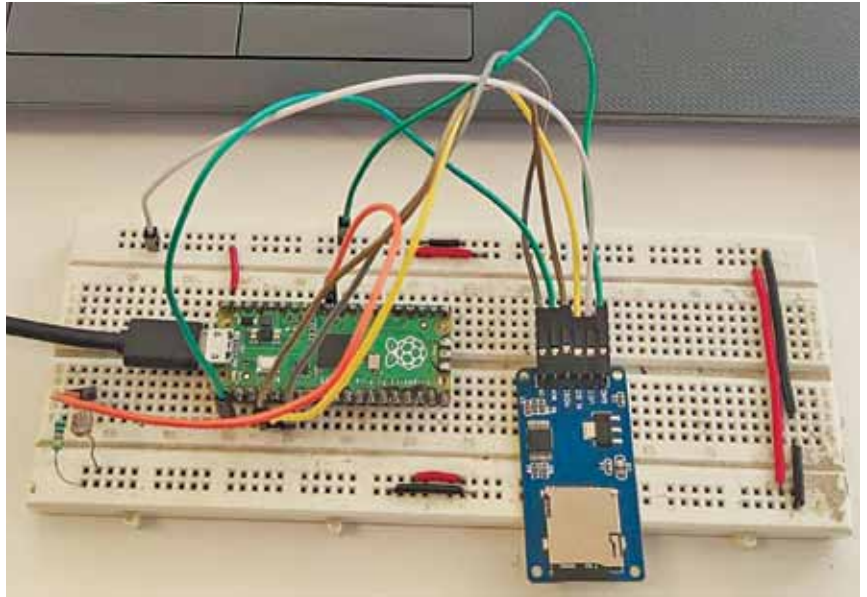


Fig. 1: Author's prototype

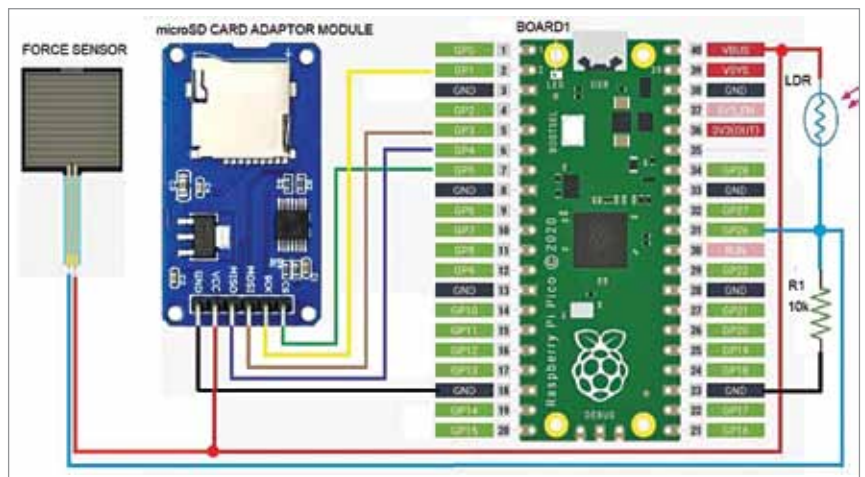


Fig. 2: Connection of the data logging system with microSD card and Raspberry Pi Pico

COMPONENTS REQUIRED		
Components	Quantity	Purpose
Raspberry Pi Pico	1	Computing
microSD Card (16GB) with adaptor	1	Data storage
LDR	1	External sensor
Force sensor	1	External sensor
Resistor (10-kilo-ohm)	1	Current limiter
Jumper wires	9	Connections

Board1. External sensor (light sensor, light dependent resistor, or force sensor) is connected to GP26 pin of Board1.

LDR sensor is used to record light condition in the area where it

is installed. The microSD card module is interfaced with Raspberry Pi Pico board pins GP1, GP3, GP4, and GP5. VBUS from the board is used as power supply for microSD card and LDR. A 10k pull-down